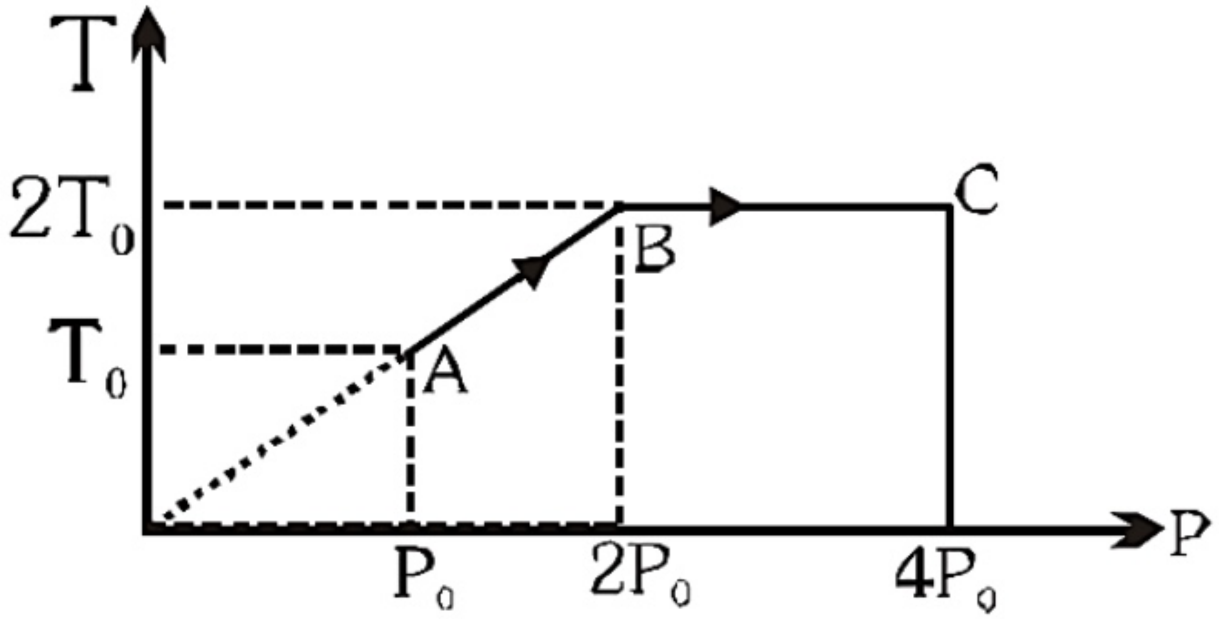
CLASS-12TH-ALL-CENTRES — with answer key

75 questions · 180 minutes · marking +4 / -1 / 0

Q1.

Single correct

One mole of an ideal gas is taken through the process ABC as shown in the figure. The total work done on the gas is:



(A) zero

(B) $2RT_0 \ln 2$ ✓

(C) $-2RT_0 \ln 2$

(D) $4RT_0 \ln 2$

Q2. Single correct

2 moles of an ideal monoatomic gas occupying volume V is adiabatically expanded from temperature $300\mathrm{K}$ to a volume of $2\sqrt{2}\,V$. Then the final temperature & change in internal energy are respectively $(R = 8.3)$

(A) $150\ \mathrm{K},\ -3735\ \mathrm{J}$ ✓

(B) $140\ \mathrm{K},\ -3735\ \mathrm{J}$

(C) $150\ \mathrm{K},\ -3537\ \mathrm{J}$

(D) $140\ \mathrm{K},\ -3537\ \mathrm{J}$

Q3. Single correct

The molar heat capacity of a monoatomic ideal gas undergoing the process $PV^{1/2} = \mathrm{constant}$ is

(A) $\frac{15}{2}R$

(B) $\frac{23}{2}R$

(C) $\frac{7}{2}R$ ✓

(D) Zero

Q4. Single correct

A gas is found to obey $P^2V = \mathrm{constant}$. The initial temperature and volume of gas are T_0 & V_0. If the gas expands to volume $3V_0$. Find the final temperature of gas.

(A) T_0

(B) $\sqrt{3}T_0$ ✓

(C) $2T_0$

(D) $\frac{T_0}{2}$

Q5. Single correct

A mixture of ideal gasses N_2 and He are taken in the mass ratio of 14 : 1 respectively. Molar heat capacity of the mixture at constant pressure is :-

(A) $\frac{19R}{6}$ ✓

(B) $\frac{6R}{19}$

(C) $\frac{13R}{6}$

(D) $\frac{6R}{13}$

Q6. Single correct

If an ideal flask containing hot coffee is shaken, the temperature of the coffee will :-

(A) decrease ✓

(B) increase

(C) remain same

(D) decrease if temperature is below 4°C and increase if temperature is equal to or more than 4°C

Q7. Single correct

Temperature of mixture of 2 moles of an ideal diatomic gas and 3 moles of an ideal monoatomic gas is increased by $3^\circ C$ at constant pressure. What is the work done by the gas mixture?
$(R \text{ is universal gas constant})$

(A) $18R$

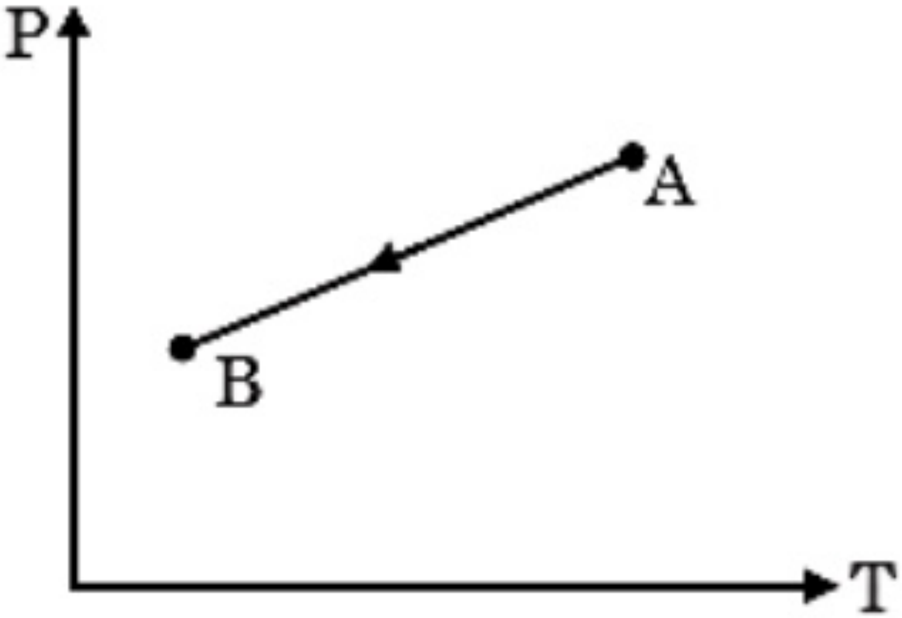
(B) $24R$

(C) $15R$ ✓

(D) $27R$

Q8. Single correct

An ideal gas undergoes a process $A \rightarrow B$ as shown on the pressure-temperature graph. The gas undergoes :-



(A) Compression ✓

(B) Expansion

(C) No change in volume

(D) Data insufficient

Q9. Single correct

A force acting on a particle moving in the xy-plane is given by $\vec{F} = (2y\hat{i} + x^2\hat{j})\mathrm{N}$, where x and y are in m. The particle moves along a straight line from the origin to $(5, 5)$. The work done by $\vec{F}$ is :

(A) $125\ \mathrm{J}$

(B) $66.7\ \mathrm{J}$ ✓

(C) $35\ \mathrm{J}$

(D) $25\ \mathrm{J}$

Q10. Single correct

Select the INCORRECT statement :-

(A) Work done by static friction on an object may be positive.

(B) Work done by a force depends on frame of reference.

(C) Kinetic friction may perform positive work on an object.

(D) Work done by gravitational force depend on path followed by object. ✓

Q11. Single correct

A body of mass $10\ \mathrm{kg}$ is released from a tower of height $20\mathrm{m}$ and body acquires a velocity of $10\mathrm{ms}^{-1}$ after falling through the distance $20\mathrm{m}$. The work done by the push of the air on the body is:- $(\text{Take } g = 10\ \mathrm{m/s}^2)$

(A) $1500\ \mathrm{J}$

(B) $1800\ \mathrm{J}$

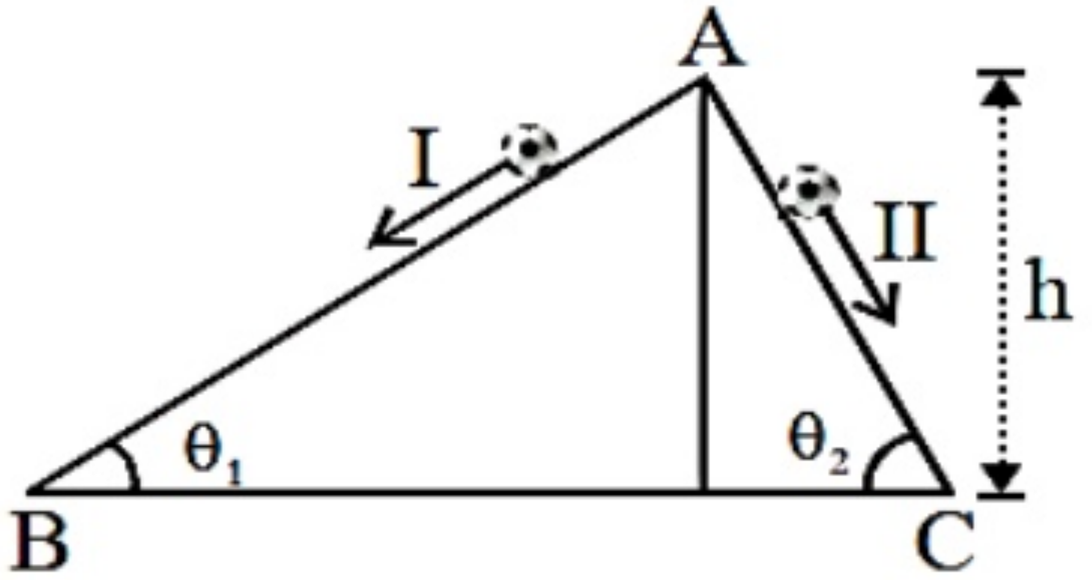
(C) $-1500\ \mathrm{J}$ ✓

(D) $-1800\ \mathrm{J}$

Q12.

Single correct

Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the following statement is correct?



(A) Both the stones reach the bottom at the same time but not with the same speed.

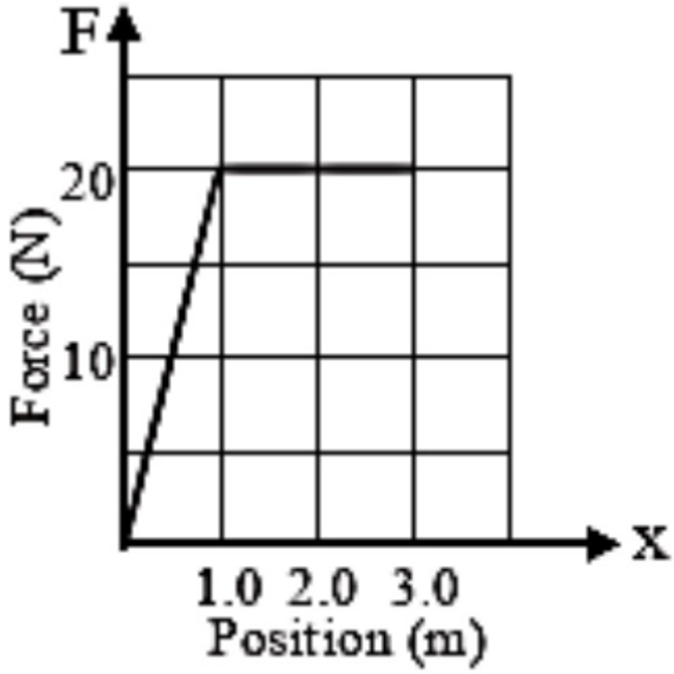
(B) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.

(C) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I. ✓

(D) Both the stones reach the bottom at different times and with different speeds.

Q13. Single correct

The graph below shows how the force on a mass depends on the position of the mass. What is the change in the kinetic energy of the mass as it moves from $x = 0.0\ \mathrm{m}$ to $x = 3.0\ \mathrm{m}$?



(A) $0.0\ \mathrm{J}$

(B) $20\ \mathrm{J}$

(C) $50\ \mathrm{J}$ ✓

(D) $60\ \mathrm{J}$

Q14. Single correct

Potential energy of a particle is $U = \frac{x^3}{3} - \frac{5x^2}{2} + 4x - 3$. If it is not experiencing any other force, then find position where it is in stable equilibrium.

(A) $x = 1$

(B) $x = -1$

(C) $x = 4$ ✓

(D) none

Q15. Single correct

A toy car can deliver a constant power of $20\mathrm{W}$. The resistive force on the car is αv, where v is velocity of car in m/s. If maximum velocity of car is $2\mathrm{m}/\mathrm{s}$, the value of α is.

(A) 10

(B) 5 ✓

(C) 15

(D) None of these

Q16. Single correct

A man is standing in the middle of a perfectly smooth ‘island of ice’ where there is no friction between the ground and his feet. Under these circumstances

(A) he can reach the desired corner by throwing any object in the same direction

(B) he can reach the desired corner by throwing any object in the opposite direction ✓

(C) he has no chance of reaching any corner of the island

(D) he can reach the desired corner by pursuing on the ground in that direction

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Q17. Single correct

A block of mass $20\ \mathrm{kg}$ is pushed with a horizontal force of $90\mathrm{N}$. If the coefficient of static and kinetic friction are 0.4 and 0.3, the frictional force acting on the block is $\left(g=10\mathrm{ms}^{-2}\right)$

(A) $90\mathrm{N}$

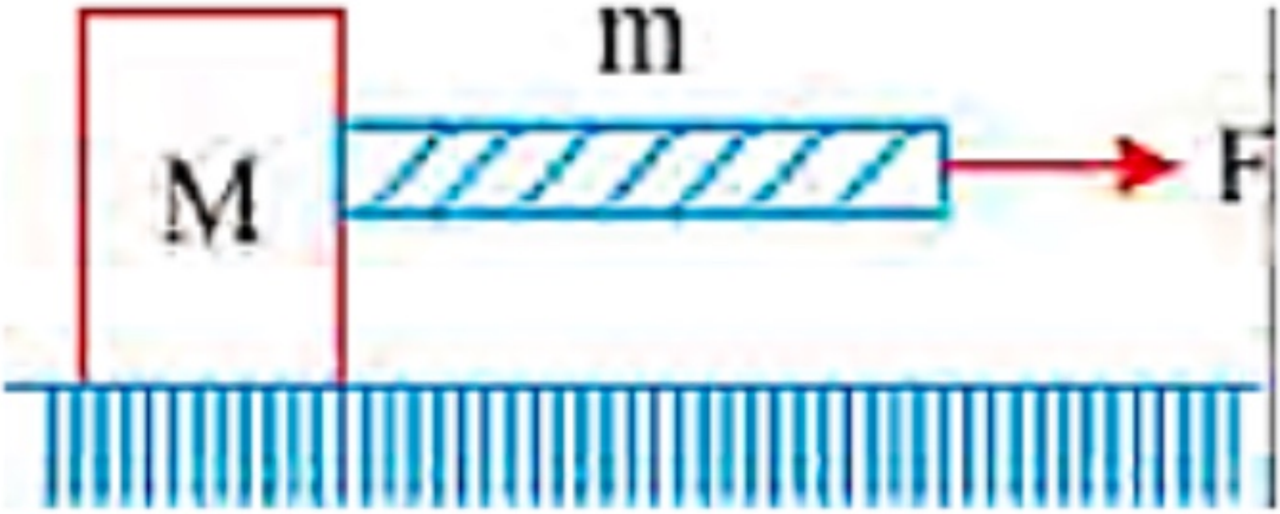
(B) $80\mathrm{N}$

(C) $60\mathrm{N}$ ✓

(D) $30\mathrm{N}$

Q18. Single correct

The block is placed on a frictionless surface in gravity free space. A heavy string of a mass m is connected and force F is applied on the string, then the tension at the middle of rope is



(A) $\frac{\left(\frac{m}{2}+M\right)F}{m+M}$ ✓

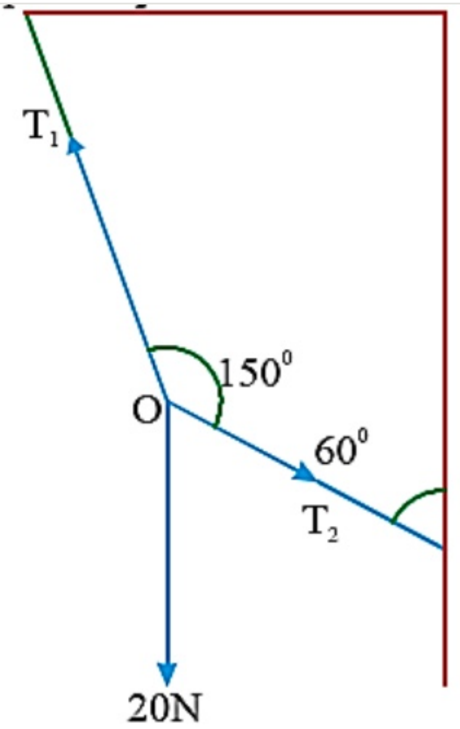
(B) $\frac{\left(\frac{M}{2}+m\right)F}{m+M}$

(C) zero

(D) $\frac{MF}{m+M}$

Q19. Single correct

If 'O' is at equilibrium then the values of the tension T_1 and T_2, respectively.



(A) $20\text{N},\ 30\text{N}$

(B) $20\sqrt{3}\text{N},\ 20\text{N}$ ✓

(C) $20\sqrt{3}\text{N},\ 20\sqrt{3}\text{N}$

(D) $10\text{N},\ 30\text{N}$

Q20. Single correct

The minimum force required to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2. If the inclined plane makes an angle θ with the horizontal such that $\tan\theta = 2\mu$, then the ratio $\frac{F_1}{F_2}$ is

(A) 4

(B) 1

(C) 2

(D) 3 ✓

Q21. Numerical

A diatomic gas $(\gamma = 1.4)$ does $2000\ \mathrm{J}$ of work when it is expanded isobarically. Find the heat given to the gas in the above process (in kJ).

Answer: 7

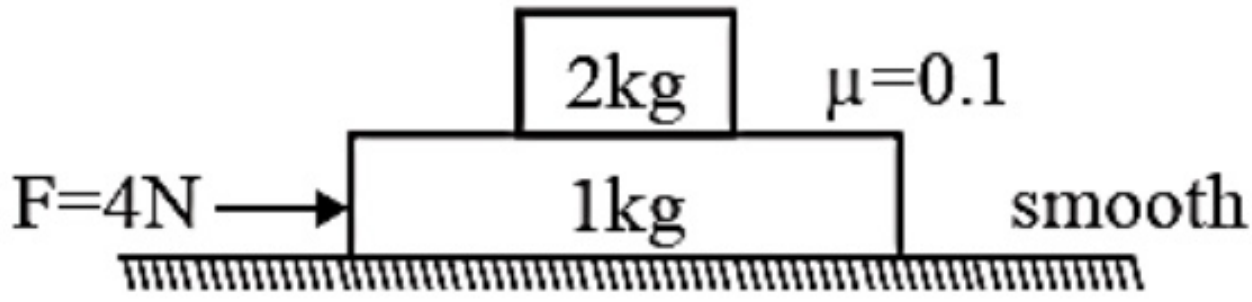
Q22. Numerical

To heat a certain amount of gas by $50^\circ\mathrm{C}$ at constant pressure, 160 calorie is required. If the same gas is heated by $100^\circ\mathrm{C}$ at constant volume, 240 calorie is required. Degree of freedom of the molecule of the gas is ________.

Answer: 6

Q23. Numerical

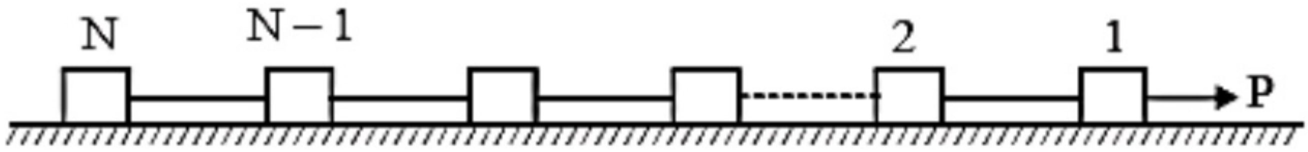
2 kg block is kept on 1 kg block as shown. The friction between 1 kg block and fixed surface is absent and the coefficient of friction between 2 kg block and 1 kg block is $\mu = 0.1$. A constant horizontal force $F = 4\ \mathrm{N}$ is applied on 1 kg block. If the work done by the friction on 1 kg block in 2s is $-X\mathrm{J}$, then find the value of X.



Answer: 8

Q24. Numerical

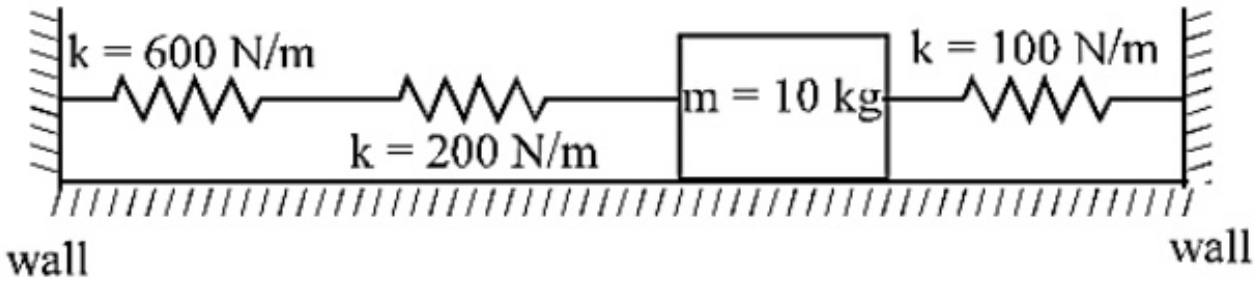
Figure shows N identical blocks connected with identical strings on a smooth horizontal surface. A constant force is pulling the blocks horizontally. If tension in string connecting 4th and 5th block is two times that between 8th and 9th block what is total number of blocks.



Answer: 12

Q25. Numerical

A 10 kg mass is stationary in the position shown. Each of the three springs are relaxed. Each spring has a characteristic stiffness constant, k, as shown on the diagram. If the mass is now displaced 20 mm to the right, what will be the net force (in newtons) exerted by the three springs on the mass?



Answer: 5

Q26. Single correct

In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen ?

(A) $2 \rightarrow 5$

(B) $3 \rightarrow 2$

(C) $5 \rightarrow 2$ ✓

(D) $4 \rightarrow 1$

Q27. Single correct

Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 \mathrm{~m~s}^{-1}$ (Mass of proton = 1.67×10^{-27} kg and h = 6.63×10^{-34} J s) :

(A) 2.5 nm

(B) 14.0 nm

(C) 0.032 nm

(D) 0.40 nm ✓

Q28. Single correct

The de Broglie wavelength of a car of mass 1000 kg and velocity 36 km/hr is :
$h = 6.63 \times 10^{-34}\ \mathrm{J\ s}$

(A) $6.626 \times 10^{-31}\ \mathrm{m}$

(B) $6.626 \times 10^{-34}\ \mathrm{m}$

(C) $6.626 \times 10^{-38}\ \mathrm{m}$ ✓

(D) $6.626 \times 10^{-30}\ \mathrm{m}$

Q29. Single correct

Which of the following is the energy of a possible excited state of hydrogen?

(A) $-3.4\ \mathrm{eV}$ ✓

(B) $+6.8\ \mathrm{eV}$

(C) $+13.6\ \mathrm{eV}$

(D) $-6.8\ \mathrm{eV}$

Q30. Single correct

Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect ? (The Bohr radius is represented by a_0)

(A) The electron can be found at a distance $2a_0$ from the nucleus

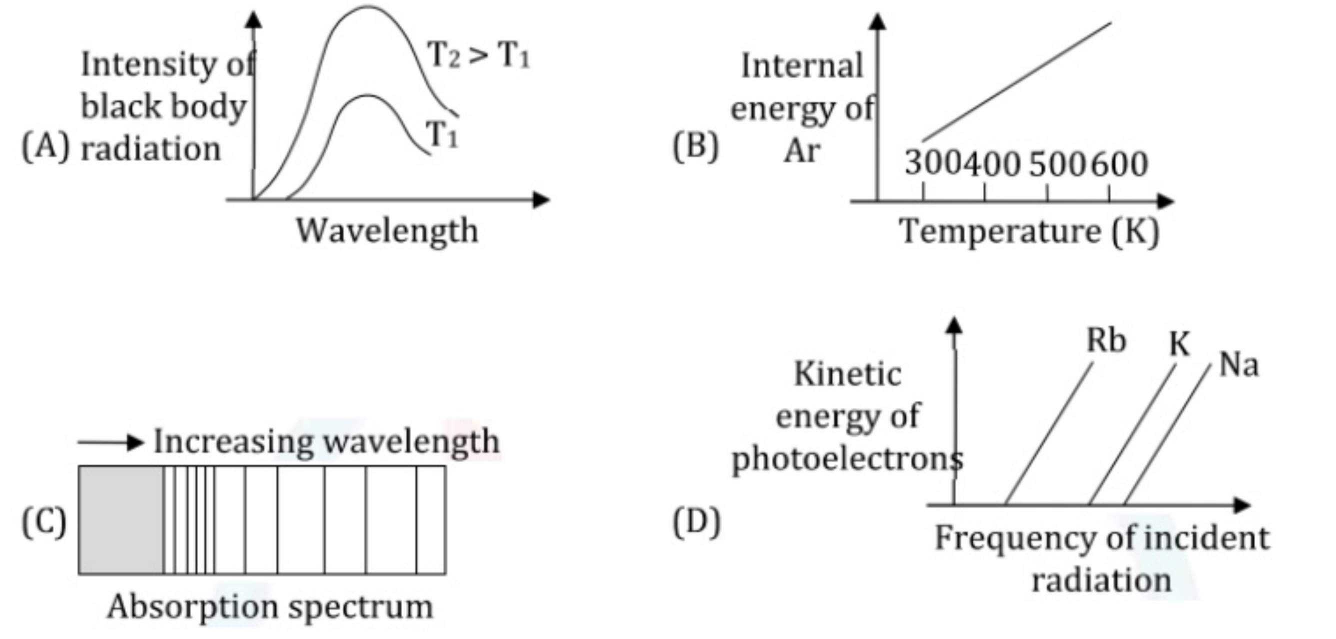
(B) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus. ✓

(C) The magnitude of potential energy is double that of its kinetic energy on an average.

(D) The probability density of finding the electron is maximum at the nucleus.

Q31. Single correct

The figure that is not a direct manifestation of the quantum nature of atoms is:



(A) A

(B) B ✓

(C) C

(D) D

Q32. Single correct

The correct statement about probability density (except at infinite distance from nucleus) is :

(A) It can negative for 2p orbital

(B) It can be zero for 1s orbital

(C) It can never be zero for 2s orbital

(D) It can be zero for 3p orbital ✓

Q33. Single correct

Which one of the following statement is False ?

(A) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction

(B) The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$ where M is the molarity of the solution

(C) The correct order of osmotic pressure for 0.01M aqueous solution of each compound is $\mathrm{BaCl} * 2 > \mathrm{KCl} > \mathrm{CH} * 3\mathrm{COOH} > \mathrm{Sucrose}$

(D) Two sucrose solutions of same molality prepared in different solvent will have the same freezing point depression ✓

Q34. Single correct

A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be

(A) 350 ✓

(B) 300

(C) 700

(D) 360

Q35. Single correct

K_f for water is $1.86 \mathrm{~K} \mathrm{~kg} \mathrm{~mol}^{-1}$. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol $[\mathrm{C} * 2\mathrm{H} * 6\mathrm{O} * 2]$ must you add to get the freezing point of the solution lowered to −2.8°C ?

(A) 27 g

(B) 72 g

(C) 93 g ✓

(D) 39 g

Q36. Single correct

The vapour pressure of acetone at 20°C is 185 torr. When 1.2 g of non-volatile substance was dissolved in 100 g of acetone at 20°C, its vapour pressure was 183 torr. The molar mass $(\mathrm{g} \mathrm{~mol}^{-1})$ of the substance is :

(A) 128

(B) 488

(C) 32

(D) 64 ✓

Q37. Single correct

Which one of the following statements regarding Henry's law not correct ?

(A) The value of K_H increases with function of the nature of the gas

(B) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids. ✓

(C) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.

(D) Different gases have different K_H (Henry's law constant) values at the same temperature.

Q38. Single correct

The vapour pressures of pure liquids A and B are 400 and 600 mmHg, respectively at 298 K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are :

(A) 450 mmHg, 0.4, 0.6

(B) 500 mmHg, 0.4, 0.6 ✓

(C) 500 mmHg, 0.5, 0.5

(D) 450 mmHg, 0.5, 0.5

Q39. Single correct

A solution is prepared by dissolving 0.6 g of urea (molar mass = 60 g mol^{-1}) and 1.8 g of glucose (Molar mass = 180 g mol^{-1}) in 100 mL of water at 27°C. The osmotic pressure of the solution is:
$R = 0.08206 \text{ L atm K}^{-1} \text{ mol}^{-1}$

(A) 4.92 atm ✓

(B) 2.46 atm

(C) 8.2 atm

(D) 1.64 atm

Q40. Single correct

Consider an elementary reaction
$\mathrm{A}(g)+\mathrm{B}(g) \rightarrow \mathrm{C}(g)+\mathrm{D}(g)$\nIf the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become ' x ' times of the original reaction rate. The value of x is:

(A) 3

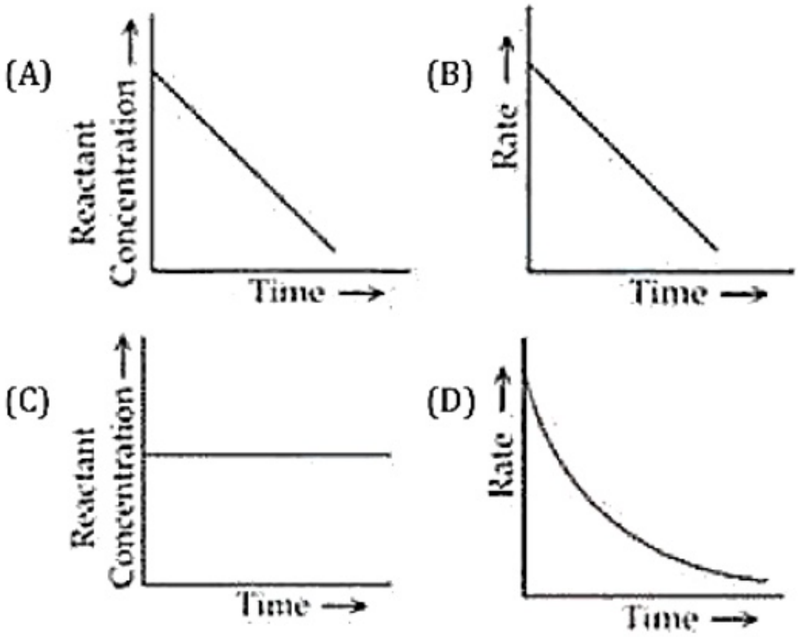
(B) $\frac{1}{9}$

(C) 9 ✓

(D) $\frac{1}{3}$

Q41. Single correct

Which of the following graphs most appropriately represents a zero order reaction?



(A) A ✓

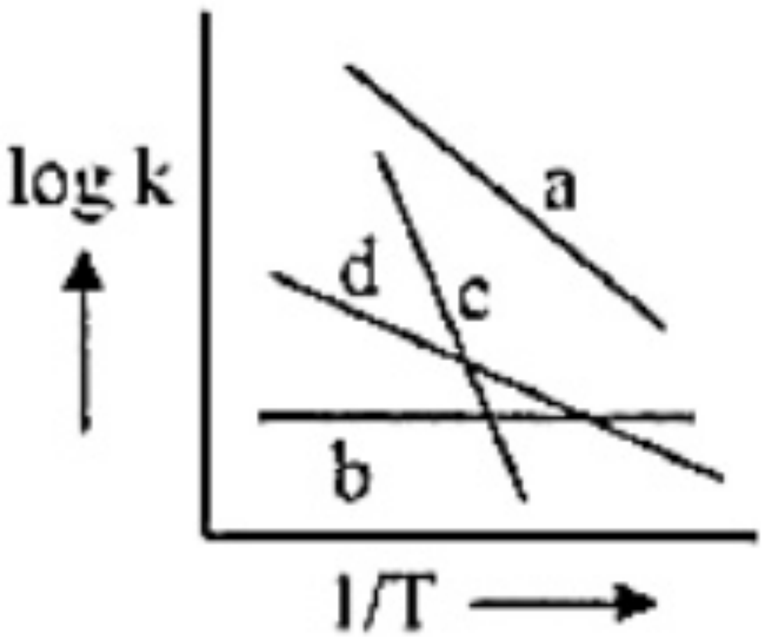
(B) B

(C) C

(D) D

Q42. Single correct

Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is corrects for the activation energies of these reactions?



(A) $E_a > E_c > E_d > E_b$

(B) $E_b > E_a > E_d > E_c$

(C) $E_c > E_a > E_d > E_b$ ✓

(D) $E_b > E_d > E_c > E_a$

Q43. Single correct

For an elementary chemical reaction, $\mathrm{A} * 2 \to 2\mathrm{A}$, the expression for $\frac{d[\mathrm{A}]}{dt}$ is:

(A) $2k_1[\mathrm{A} * 2] - 2k * -1[\mathrm{A}]^2$

(B) $k_1[\mathrm{A} * 2] - k * -1[\mathrm{A}]^2$

(C) $2k_1[\mathrm{A} * 2] - k * -1[\mathrm{A}]^2$ ✓

(D) $k_1[\mathrm{A} * 2] + k * -1[\mathrm{A}]^2$

Q44. Single correct

Higher order (>3) reactions are rare due to :-

(A) shifting of equilibrium towards reactants due to elastic collision

(B) loss of active species on collision

(C) low probability of simultaneous collision of all the reacting species ✓

(D) increase in entropy and activation energy as more molecules are involved.

Q45. Single correct

A reactant (A) forms two products :

$\mathrm{A} \xrightarrow{k_1} \mathrm{B},\ \text{Activation Energy } E * a1$

$\mathrm{A} \xrightarrow{k_1} \mathrm{C},\ \text{Activation Energy } E * a2$\nIf $E * a2 = 2E * a1$, then k_1 and k_2 are related as :-

(A) $k_1 = 2k_2 e^{E*a2/RT}$

(B) $k_1 = k_2 e^{E*a1/RT}$

(C) $k_2 = k_1 e^{E_{a2}/RT}$

(D) $k_1 = Ak_2 e^{E_{a1}/RT}$ ✓

Q46. Numerical

The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is ***. (Nearest Integer)
(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

Answer: 1758

Q47. Numerical

If wavelength of the first line of the Paschen series of hydrogen atom is 720 nm, then the wavelength of the second line of this series is***___ nm. (Nearest integer)

Answer: 492

Q48. Numerical

1 kg of 0.75 molal aqueous solution of sucrose is cooled to −4°. The amount of ice (in g) that will be separated out is ______ . (Nearest integer)
[Given : $K_f(\mathrm{H} * 2\mathrm{O}) = 1.86 \mathrm{~K~kg~mol}^{-1}$]

Answer: 518

Q49. Numerical

A 1 molal $\mathrm{K} * 4\mathrm{Fe}(\mathrm{CN}) * 6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another solution which contains 18.1 weight percent of a non electrolyte solute A. The molar mass of A is***_ u.
(Round off to the Nearest Integer).
[Density of water = $1.0 \mathrm{~g~cm}^{-3}$]

Answer: 85

Q50. Numerical

The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is _____ $\times 10^{-5} \mathrm{~mol~dm}^{-3}$.
(Round off to the Nearest Integer)
[Given : Henry's law constant = $K_H = 8.0 \times 10^4 \mathrm{~kPa}$ for O_2.\nDensity of water with dissolved oxygen = $1.0 \mathrm{~kg~dm}^{-3}$]

Answer: 25

Q51. Single correct

$\lim\limits_{x \to \infty} \dfrac{\log x^n - [x]}{[x]}$, $n \in N$, (where $[x]$ denotes greatest integer less than or equal to x)

(A) Has value -1 ✓

(B) Has value 0

(C) Has value 1

(D) Does not exist

Q52. Single correct

Let $f:(1,\infty)\rightarrow(0,\infty)$ be a continuous decreasing function with $\lim_{x\to\infty}\frac{f(4x)}{f(8x)}=1$. Then $\lim_{x\to\infty}\frac{f(6x)}{f(8x)}$ is equal to

(A) $\frac{4}{8}$

(B) $\frac{4}{6}$

(C) $\frac{6}{8}$

(D) 1 ✓

Q53. Single correct

If $f(x)=xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}$ for $x\neq 0$, and $f(x)=0$ for $x=0$, then f(x) is

(A) discontinuous everywhere

(B) continuous as well as differentiable for all x

(C) continuous for all x but not differentiable at $x=0$ ✓

(D) neither differentiable nor continuous at $x=0$

Q54. Single correct

The value of the limit $\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)$ is

(A) 1

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$ ✓

Q55. Single correct

Let $f(x)=\left[\frac{\sin x}{x}\right]+\left[\frac{2\sin 2x}{x}\right]+\ldots+\left[\frac{10\sin 10x}{x}\right]$ (where $[y]$ is the largest integer $\leq y$). The value of $\lim_{x\to 0} f(x)$ equals

(A) 55

(B) 164

(C) 165

(D) 375 ✓

Q56. Single correct

$\lim_{n\to\infty}\frac{1^2n+2^2(n-1)+3^2(n-2)+\ldots+n^2\cdot 1}{1^3+2^3+3^3+\ldots+n^3}$ is equal to

(A) $\frac{1}{3}$ ✓

(B) $\frac{2}{3}$

(C) $\frac{1}{2}$

(D) $\frac{1}{6}$

Q57. Single correct

Let function g(x) be differentiable and $g'(x)$ is continuous in $(-\infty,\infty)$ with $g'(2)=14$, then $\lim_{x\to 0}\frac{g(2+\sin x)-g(2+x\cos x)}{x-\sin x}$ is equal to

(A) 7 ✓

(B) 14

(C) 28

(D) 56

Q58. Single correct

Let $f:(0,\infty)\to(-\infty,\infty)$ be defined as $f(x)=e^x+\ln x$ and $g=f^{-1}$, then

(A) $g''(e)=\frac{1-e}{(1+e)^3}$

(B) $g''(e)=\frac{e-1}{(1+e)^3}$

(C) $g'(e)=e+1$ ✓

(D) None of these

Q59. Single correct

If $\lim\limits_{x\to 0}\dfrac{1-\sqrt{\cos 2x}\cdot\sqrt[3]{\cos 3x}\cdot\sqrt[4]{\cos 4x}\ldots\sqrt[n]{\cos nx}}{x^2}$ has the value equal to 10 then the value of n equals

(A) 4

(B) 5

(C) 6 ✓

(D) 7

Q60. Single correct

If $2x+y=1$ be a tangent to $y=f(x)$ at $x=\frac{1}{3}$ then $\lim\limits_{x\to 0}\dfrac{\sin x(\sin x-1)}{f\left(\frac{e^{3x}}{3}\right)-f\left(\frac{e^{-3x}}{3}\right)}$ is equal to

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$ ✓

(D) $-\frac{1}{4}$

Q61. Single correct

Let $f : R \to R$ be a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3),\ x \in R$. Then $f(2)$ equals

(A) 30

(B) -2 ✓

(C) -4

(D) 8

Q62. Single correct

Let $f : R \to R$ be a differentiable function satisfying $f'(3) + f'(2) = 0$. Then $\lim\limits_{x \to 0} \left(\dfrac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}}$ is equal to

(A) e

(B) e^{-1}

(C) 1 ✓

(D) e^2

Q63. Single correct

Let $f : R \to R$, $g : R \to R$ and $h : R \to R$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in R$. Then

(A) $g'(2) = \frac{1}{15}$

(B) $h'(1) = 666$ ✓

(C) $h(1) = 16$

(D) $h(g(3)) = 36$

Q64. Single correct

If $f(x) = xe^{x(1-x)}$, then f(x) is

(A) increasing on $\left(-\frac{1}{2}, 1\right)$ ✓

(B) decreasing on $\left[-\frac{1}{2}, 1\right]$

(C) increasing on R

(D) decreasing on R

Q65. Single correct

If f is a strictly increasing function, then $\lim_{x \to 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is equal to

(A) 0

(B) 1

(C) -1 ✓

(D) 2

Q66. Single correct

Let $y = f(x)$ be drawn with $f(0) = 2$ and for each real number 'a' the line tangent to $y = f(x)$ at $(a, f(a))$, has x-intercept $(a - 2)$. If $f(x)$ is of the form of ke^{px}, then $\left(\frac{k}{p}\right)$ has the value equal to

(A) 4 ✓

(B) 3

(C) 2

(D) 1

Q67. Single correct

Let $f(x) = 2 - |x^2 + 5x + 6|$ for $x \neq -2$, and $f(-2) = b^2 + 1$. If f(x) has relative maximum at $x = -2$, then the range of the b, is

(A) $|b| \geq 1$ ✓

(B) $|b| < 1$

(C) $b > 1$

(D) $b < 1$

Q68. Single correct

Let f and g be two differentiable functions defined from $R \to R^+$. If f(x) has a local maximum at $x = c$ and g(x) has a local minimum at $x = c$, then $h(x) = \frac{f(x)}{g(x)}$

(A) has a local maximum at $x = c$ ✓

(B) has a local minimum at $x = c$

(C) is monotonic at $x = c$

(D) has a point of inflection at $x = c$

Q69. Single correct

$f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5,\ x > 1$, which one of the following is NOT correct ?

(A) f is increasing in $(1, 2)$ and decreasing in $(2, \infty)$

(B) $f(x) = -1$ has exactly two solutions

(C) $f'(e) - f''(2) < 0$ ✓

(D) $f(x) = 0$ has a root in the interval $(e, e+1)$

Q70. Single correct

Let the function $f(x) = 2x^3 + (2p-7)x^2 + 3(2p-9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is:

(A) $\left(\frac{9}{2}, \infty\right)$

(B) $\left(0, \frac{9}{2}\right)$

(C) $\left(-\infty, \frac{9}{2}\right)$ ✓

(D) $\left(-\frac{9}{2}, \frac{9}{2}\right)$

Q71. Numerical

Let $[t]$ denote the greatest integer $\leq t$ and $\{t\}$ denote the fractional part of t. Then integral value of a for which the left hand limit of the function $f(x) = [1+x] + \dfrac{\alpha^{2[x]+\{x\}} + [x] - 1}{2[x] + \{x\}}$ at $x = 0$ is equal to $\alpha - \frac{4}{3}$ is ____

Answer: 3

Q72. Numerical

Let $f : R \to R$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in R$. Then, the value of $\log_e(f(4))$ is ______.

Answer: 2

Q73. Numerical

Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to

Answer: 19

Q74. Numerical

Let f, g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(fg)'(0) = 6$; $(gh)'(0) = 4$ and $(hf)'(0) = 5$ then compute the value of $(fgh)'(0)$.

Answer: 16

Q75. Numerical

Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower-bed, is

Answer: 25
